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Shared CO_2 capture, transport, and storage for decarbonizing industrial clusters

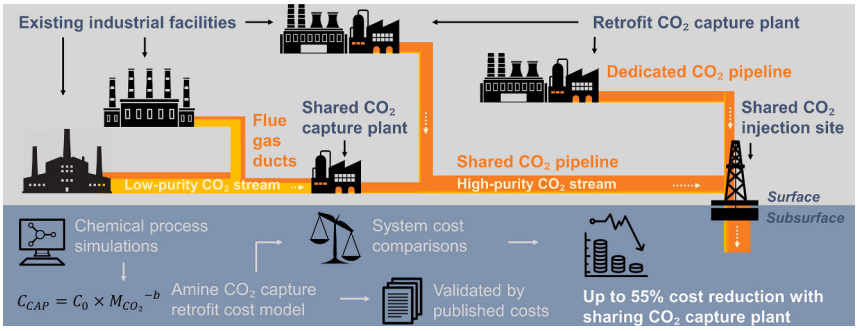
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HIGHLIGHTS

- Amine CO_2 capture retrofit cost model developed for any industrial facility.
- Capture cost model validated against published costs for actual retrofit projects.
- Capture plants shared by multiple facilities reduces average capture costs.
- Shared CO_2 transport dramatically reduces required pipeline mileage and costs.
- Cost benefits of shared infrastructure are most significant for smaller facilities.

GRAPHICAL ABSTRACT



ARTICLE INFO

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ABSTRACT

A model for estimating CO_2 capture retrofit costs at many types of industrial facilities is developed and then applied in a case study exploring alternative designs for capture, transport, and underground storage of CO_2 from a cluster of industrial facilities in Southeast Louisiana, USA. The capture cost model is anchored by granular chemical process simulations used to determine capacities of individual equipment components, the capital costs for which are estimated using factoring methods. To generalize the cost model, process simulations are developed for target capture streams having CO_2 concentrations of 5, 10, 15, and 94 mol%, and for each concentration, seven different scales of capture plants are modeled. The cost model is then embedded in *SimCCS*^{PRO}, a customized version of open-source software for optimizing CO_2 pipeline capacities and routings to underground storage sites. For a 22-facility cluster of industrial CO_2 sources with collective emissions of 8.1 million tCO_2/year today, we explore capture, transport and storage (CTS) system designs with varying levels of shared capture and transport infrastructure. When CO_2 pipelines are shared rather than dedicated to individual capture facilities, average transport costs can be reduced by up to two-thirds (and aggregate pipeline length by more than this) for the same level of CO_2 capture and storage. However, capture costs dominate total CTS costs. Because of this, pooling emission streams from multiple facilities and sharing the scale-economy benefits of larger capture facilities enables more significant reductions in CTS costs per tonne of CO_2 stored, even though some of the savings are offset by the added flue gas transport costs. The cost benefits of shared infrastructure are most significant for smaller facilities, i.e., with emissions <0.1 million tCO_2/year.

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Nomenclature

c	Price, $ per unit of energy, mass, or area
C	Cost, $
$CAPEX$	Capital expenditure, $
CRF	Capital recovery factor, dimensionless
d	Wall thickness, m
D	Diameter, m
HHV	Higher heating value, kWh/kg
i	Weighted average cost of capital, yr^{-1}
JF	Project contingency factor, dimensionless
LCC	Levelized cost of CO_2 capture, $/t$CO_2$
LCS	Levelized cost of CO_2 storage, $/t$CO_2$
LCT	Levelized cost of CO_2 transport, $/t$CO_2$
$LCTS$	Levelized cost of CO_2 capture, transport, and storage, $/t$CO_2$
LF	Lang factor, dimensionless
LHV	Lower heating value, kWh/kg
m	Molar mass flow rate, kmol/h
M	Molar mass, g/mol
$\dot{m}$	Mass flow rate, kg per hour
$\dot{M}$	Mass flow rate, million tonnes per year
n	Number of equivalents per mole, dimensionless
N	Number, unit
$OPEX$	Operational and maintenance expenditure, $
p	Pressure, barg
P	Power, kW_e
Q	Heat, kW_{heat}
r	Radius, km
RF	Retrofit factor, dimensionless
SF	Process contingency factor, dimensionless
T	Time, year
$TPEC$	Total purchased equipment cost, $

Greek symbols

α	Solution loading, mol/mol
η	Efficiency, %
λ	Capacity factor, %
μ	Specific energy consumption, kWh/tCO_2
π	Pi = 3.14, dimensionless
ρ	Mass density, kg/m^3
σ	Area density, ktCO_2/km^2
ψ	Percentage of CO_2 is recovered, %
ω	Mass fraction, dimensionless

Subscripts and superscripts

A&O	Administrative and general overhead
CAP	CO_2 capture plant
CC	CO_2 capture system
CO_2	Carbon dioxide
CS	CO_2 storage system
CT	CO_2 transport system
CCS	CO_2 capture and storage
CTS	CO_2 capture, transport, and storage system
COG	Cogeneration plant
COL	Cooling water plant
COM	CO_2 compression plant
DEH	CO_2 dehydration plant
ELC	Electricity
EOR	Enhanced Oil Recovery
F	Facility
FG	Flue gas
FIX	Fixed operating and maintenance costs
FOAK	1st-of-a-kind
H_2O	Water
IFT	Insurance, local taxes and fees
IN	Inner
K	Construction
LAB	Labor
LS	Lean solvent
MEA	Monoethanolamine
N_2	Nitrogen gas
NAT	Natural gas
NOAK	N^{th}-of-a-kind
O&M	Operating and maintenance
O_2	Oxygen gas
OUT	Outer
RS	Rich solvent
S	Material
USA	United States of America
VAR	Variable operating and maintenance costs
Z	Cluster

1. Introduction

Direct CO_2 emissions from the industrial sector account for about one-quarter of all greenhouse gas emissions globally [1] and in the United States [2]. With significant CO_2 abatement costs, these are often classified as hard-to-abate emissions [3]. Potential strategies for significantly decarbonizing U.S. manufacturing by 2050 include [4]: energy efficiency, raw material efficiency, feedstock and/or fuel switching to renewables, clean electrification, and CO_2 capture and storage (CCS). Potential emissions reductions with CCS are estimated to be greater than with electrification and fuel switching [5], and CCS is considered essential for some industries with limited decarbonization alternatives, such as cement [6], iron and steel [7], and chemicals [8].

In this paper, we first develop a generalized expression for estimating amine-based CO_2 capture retrofit costs for a diverse array of existing industrial facilities based on the CO_2 concentration in a facility's capture stream and its targeted capacity of the capture operation. We build thermodynamic process simulations to determine the underlying equipment capacities that then provide the basis for our cost estimations. Key elements differentiating our work from other cost modelling studies in the literature include modelling and costing of integrated CO_2 capture systems that include (1) CO_2 capture plant, (2) CO_2 compression plant, (3) CO_2 dehydration plant, (4) cogeneration plant, and (5) cooling water plant. Each integrated CO_2 capture system that we model is designed for a different combination of input target stream flow rate and CO_2 concentration. We model a range of flow rates and concentrations in order to develop a capture cost model that can be used to estimate capture costs for individual capture facilities across a wide range of sizes and industrial facility types, as well as for CO_2 capture systems that remove CO_2 from a gas stream consisting of combined flue gases from several different facilities. We incorporate our cost model into *SimCCS* [9], an open-source computational tool for optimizing the design of CO_2 pipelines linking CO_2 capture points with underground storage sites.

We then carry out assessments of alternative designs for sharing of CO_2 capture, transport, and storage infrastructure among clusters of diverse industrial facilities, including arrangements where flue gases are ducted to a centralized capture plant. We focus especially on potential benefits for small-to-medium sized emitters, which have not been the focus of most prior studies. Potential benefits of shared infrastructure include shared costs and risks, reduced average costs per tonne of CO_2 sequestrated, and reduced pipeline miles and number of injection sites [10]. Our analysis adopts a case-study approach, focusing on an existing

industrial corridor in Southeast Louisiana.

2. Prior work

Removal of CO_2 from industrial gas streams has been practiced for decades. In the U.S. the installed capacity for CO_2 removal at natural gas processing plants was about 0.4 million tonnes per year ($MtCO_2/y$) in 1972 [11,12] and grew to about 25 $MtCO_2/y$ by 2019, when it accounted for an estimated 63% of global CO_2 removal capacity [13]. Among different CO_2 removal technologies as described by Feron et al. [14], Chao et al. [15], and National Energy Technology Laboratory's (NETL) publications [16,17], absorption systems using amine solvents are the most technologically mature and widely utilized for removing CO_2 from the near-atmospheric-pressure streams with relatively low CO_2 concentrations [10,12,18,19] commonly found at most existing industrial facilities. State-of-the-art amine-based systems can remove 95% or more of the CO_2 in an emissions stream [18].

Costs for CO_2 capture to decarbonize fossil-fuel power plants have been extensively modeled by Rubin et al. [20], Vasudevan et al. [21], and NETL [22,23]. However, modelling for industrial facilities is more limited. Some studies have utilized simplified (technology-agnostic) cost estimation methods, for example Bains et al. [24] for petroleum power plant, natural gas power plant, coal power plant, cement, refinery, iron and steel, ethylene, ethylene oxide, ammonia, hydrogen, ethanol and natural gas processing, and Pilorgé et al. [25] for process-originated emissions in refining, hydrogen, petrochemicals, ammonia, bio-ethanol, iron and steel, cement and lime. Other studies include more-detailed technology modelling, but for limited types of industries e.g., Leeson et al. [26] for cement, refinery, iron and steel, pulp and paper, ethylene oxide, ammonia, hydrogen, ethanol and natural gas processing, Lisbona et al. [27] for cement and some generic (unspecified) coal or natural gas fueled industrial facilities, and Hughes and Zoelle [28] for ammonia, cement, iron and steel, ethylene oxide, ethanol, hydrogen, natural gas processing, coal-to-liquids, and gas-to-liquids. Moreover, industrial analyses by Tock et al. [29] and Pérez-Calvo et al. [30], including otherwise excellent studies do not include additional energy generation (and its associated costs) for meeting parasitic heat and electricity needs of the CO_2 capture plant. The cost for such energy supply is important, since even when some waste heat is available from the existing industrial process, it will typically be insufficient as described by Johnsson et al. [31] and Roussanaly et al. [32]. Our carbon capture modelling work here addresses these limitations of prior studies. In particular, this study explores CO_2 capture for retrofitting at a wide range of existing industrial facilities, since retrofits will be important for achieving emissions reductions rapidly [10,33].

Beyond CO_2 capture cost studies, there is also attention paid in the literature to the integration of CO_2 capture with transport and underground storage systems. The limited number of power or industrial sector CCS projects that have been built to date have been designed with dedicated CO_2 pipelines connecting individual point sources of CO_2 to storage sites, including at SaskPower's Boundary Dam [34], Air Product's Port Arthur [35], Shell Quest [36] and Petra Nova [37]. But policy studies addressing future large-scale deployment of CCS envision sharing of pipeline and storage infrastructures and a number of design studies for shared-infrastructure have been published, such as by Labor Energy Partnership [38], International Energy Agency [39], United States Department of Energy [40], and Great Plains Institute [41]. For example, Edwards and Celia investigated ethanol biorefineries in the Midwestern U.S. capturing CO_2 and sharing trunk pipelines to carry CO_2 to enhanced oil recovery operations in the Permian Basin in Texas [42]. They in turn used the results to understand the level of government subsidy needed for economic viability. Tao et al. investigated sharing of a pipeline and storage network among 156 prospective fossil fuel power plants in the U.S, or equivalent to 100 GW of power capacity that would be fitted with CO_2 capture [43]. Pilorgé et al. investigated capture, transport, and storage networks for process industries, but with simplified (straight line) representation of pipelines [25]. It is important to note that all of these studies were for a specific industrial type in the US that might overlook the potential of sharing pipeline networks among a range of different industrial facilities. For multi-industrial types, Sun et al. [44] applied a screening methodology to identify potential CCS clusters at a state-wide level in Spain and found complementarities between larger and smaller capture facilities sharing transport and storage infrastructure. Berghout et al. [45,46] investigated shared-infrastructure design trade-offs for a cluster of 16 facilities in an industrial zone in the city of Rotterdam and found that significant reduction of capture costs can be achieved by sharing a centralized CO_2 capture system. In another study in the Netherlands, Knoope et al. [47] assessed the impact that uncertainty in CO_2 emission prices can have on the deployment of shared CCS infrastructure and concluded that building CCS networks can be stimulated if capture costs are low and there is cooperation among nearby power and industrial facilities. However, both Berghout et al. and Knoope et al. simplified the modelling of CO_2 transport networks, similar to the approach by Pilorgé et al. Using optimized transport routes, pipeline capacities, and costs, Middleton and Yaw [48] investigated technical uncertainties in developing CCS networks and found CO_2 injection and storage introduce the largest uncertainties in the techno-economic performance of a cluster of CO_2 capture plants sharing transport and storage infrastructure. Our study extends the literature on shared CCS infrastructure designs by including small-scale facilities and a wide range of industrial types, evaluating different scenarios for minimizing overall CCS costs, considering realistic CO_2 pipeline routings and well-characterized CO_2 injection sites.

3. CO_2 capture cost model

Our CO_2 capture cost model estimates the levelized cost of CO_2 capture as a function of CO_2 concentration in the target stream and annual CO_2 capture rate. The model was developed by designing and simulating the performance of capture facilities to determine individual equipment capacities at the capture facility. Equipment purchase costs are estimated based on these capacities and then cost factoring is applied, as discussed in detail in Section 3.3, to estimate total installed capital costs. Capital cost estimates combined with fixed and variable operating cost estimates are used to calculate levelized CO_2 capture costs.

3.1. Underlying process designs

Using the *Aspen* + chemical process modelling tool, we designed and simulated the performance of amine-based CO_2 capture facilities comprising (1) CO_2 absorption/desorption columns, (2) CO_2 compression to a pressure suitable for pipeline transport, (3) CO_2 dehydration to remove residual water, (4) an integrated cogeneration plant to deliver all of the heat and power needed by the capture plant, and (5) a cooling water system (Fig. 1). By including cogeneration within the facility, our designs assume that waste heat from the existing process is unavailable to help deliver the heat needed to operate the CO_2 capture system. We note that a dedicated cogeneration plant has been built for most commercial-scale amine-based industrial CO_2 capture retrofit projects in North America, including Boundary Dam [34,49], Quest [36], and Petra Nova [37,50]. Our cogeneration system is natural gas fired, and the CO_2 emissions from this are assumed to be vented in most cases, which reduces the net CO_2 capture at the facility. However, we also consider one case in which the cogeneration CO_2 emissions are captured (see dashed line in Fig. 1).

An industrial stream targeted for CO_2 removal can be characterized by the concentration of CO_2 in the stream and the mass flow of the stream. We developed separate process simulations for target streams having CO_2 concentrations of 5 mol%, 10 mol% and 15 mol%. We designed for a CO_2 capture ratio (ψ_{CO_2}) of 95% in each case, as calculated using Eq. 1.

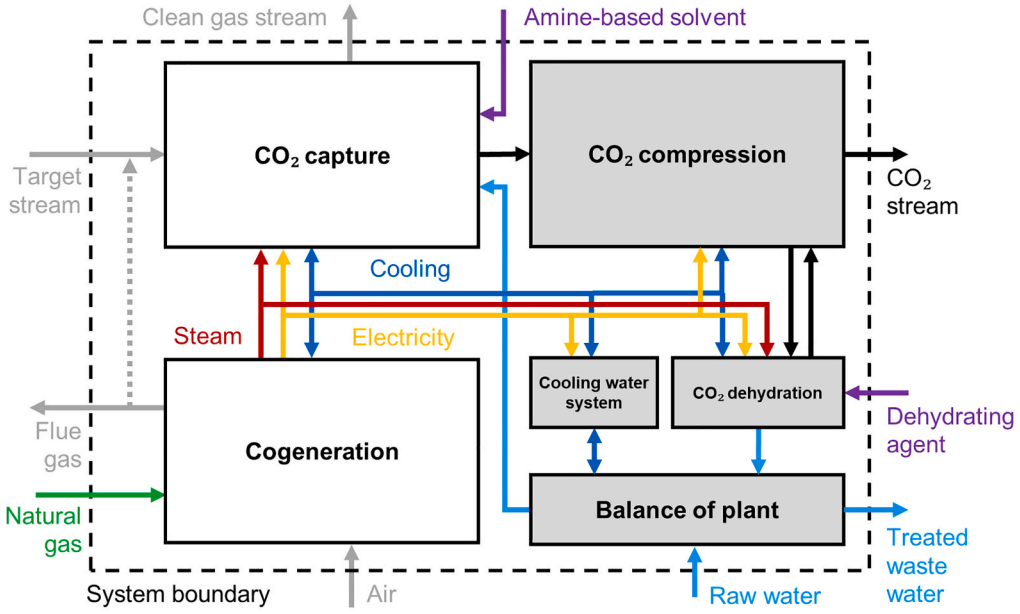


Fig. 1. Block diagram of simulated CO_2 capture facilities for processing of target streams with CO_2 concentrations of 5, 10, and 15 mol%. The cogeneration block provides all parasitic heat and electricity needs in those cases. Flue gas CO_2 emissions from cogeneration are vented to the atmosphere in some cases, and in other cases (dashed arrow) they are captured along with the target-stream CO_2. For the high-purity (94 mol%) CO_2 target streams, only the shaded blocks are included; parasitic electricity needs are supplied by the grid, and modest parasitic heat needs are met by natural gas combustion.

$$\psi_{CO_2} = \frac{\dot{m}_{CO_2,CAP}}{\dot{m}_{FG} \times \omega_{CO_2}} \tag{1}$$

where $\dot{m}_{CO_2,CAP}$ is the hourly mass flow rate of captured CO_2, $\dot{m}_{FG}$ is the target stream hourly mass flow rate, and ω_{CO_2} is the mass fraction of CO_2 in the target stream.

We further defined $\dot{M}_{CO_2,CAP}$ to be the annual capture rate ($MtCO_2/y$), if the facility were to operate at full capacity all year:

$$\dot{M}_{CO_2,CAP} = \dot{m}_{CO_2,CAP} \times 8760 \tag{2}$$

For each input CO_2 mass fraction, ω_{CO_2} we simulated capture for seven different target-stream mass flow rates, resulting in $\dot{M}_{CO_2,CAP}$ values of 0.10, 0.29, 0.57, 1.15, 2.29, 3.43, and 5.70 $MtCO_2/y$.

The predominant component in most industrial gas streams that might be targeted for CO_2 removal is nitrogen (N_2), followed by lower shares of water vapor (H_2O) and oxygen (O_2) [51,52]. For simulation purposes, we assumed target streams have H_2O and O_2 concentrations of 10 mol% and 8 mol%, respectively, and we adjust the N_2 concentration to achieve the desired CO_2 concentration (Table 1). We assume target streams arrive at the capture facility at near-atmospheric pressure, as would be commonly found in industry [19], with a temperature of 50 °C and with trace contaminant levels that do not require removal prior to CO_2 removal. Pre-treatment required for trace contaminant removal will vary widely from facility to facility, so we have opted not to model it in a generalized way, other than to acknowledge that our assumed project contingency levels (discussed in Section 3.3) subsume the cost of such pretreatment.

Table 1
Mole and mass fractions of modeled capture target streams.

Component	Mole fractions			Mass fractions		
	5 mol% CO_2	10 mol % CO_2	15 mol % CO_2	5 mol% CO_2	10 mol % CO_2	15 mol % CO_2
CO_2	0.05	0.10	0.15	0.08	0.15	0.22
N_2	0.77	0.72	0.67	0.77	0.70	0.63
O_2	0.08	0.08	0.08	0.09	0.09	0.09
H_2O	0.10	0.10	0.10	0.06	0.06	0.06

A few industrial processes generate CO_2-laden streams with very high concentrations of CO_2, e.g., ammonia production and natural gas processing plants. To represent CO_2 capture at such facilities, we developed seven additional simulations (one for each of the values of $\dot{M}_{CO_2,CAP}$ noted above) for target streams with 94 mol% CO_2. CO_2 capture systems for such high-purity streams comprise only CO_2 dehydration, compression, and associated cooling water [53], so we assume the capture ratio is 100% in these cases. These simulations utilize only the gray-shaded blocks in Fig. 1. Cogeneration is not needed because high-purity target streams require very little heat supply for capture. (See SI.12). Since no cogeneration block is included, the very small parasitic heat demand in these cases is met by a natural gas boiler (venting the resulting CO_2), and electricity needs are assumed to be supplied by the grid. Details for each block in Fig. 1 are described next. The SI.4 provides two *Aspen* + flowsheet representations of Fig. 1, one in which the CO_2 in the cogeneration flue gas is vented to the atmosphere and one in which it is captured along with CO_2 in the target stream.

The **CO_2 capture** block comprises an absorber, stripper, heat exchanger and pump. Design specifications, taken from Pellegrini et al. [54], are detailed in SI.6. Monoethanolamine (MEA) with a concentration of 30 wt% in water is adopted as the solvent due to its higher reactivity with CO_2 than other aqueous amine solvents like diethanolamine or methyldiethanolamine [16,18,29,55]. The design capture rate is 95%, considering that capture rates higher than this are likely to be uneconomic [10]. The lean solvent flow rate, $\dot{m}_{LS}$, is calculated from Eq. 3 [56]:

$$\dot{m}_{LS} = \frac{\dot{m}_{FG} \times \omega_{CO2} \times \psi_{CO2}}{100n(\alpha_{RS} - \alpha_{LS})} \left(\frac{M_{MEA}}{44.009} \left(1 + \frac{1 - \omega_{MEA}}{\omega_{MEA}} \right) + n\alpha_{LS} \right) \tag{3}$$

where M_{MEA} is the molar mass of MEA, n is the number of equivalents per mole of the amine (n is 1 for MEA), ω_{MEA} is the mass fraction of MEA in the unloaded solution, and α_{LS} and α_{RS} are the lean and rich amine solution loadings, respectively, determined from the following Eqs. [54], where m refers to molar flow rates:

$$\alpha_{LS} = \frac{m_{CO2,LS}}{m_{MEA,LS}} \tag{4}$$

$$\alpha_{RS} = \frac{m_{CO2,RS}}{m_{MEA,RS}} \quad (5)$$

The capture block is designed to maximize shop fabrication over field fabrication to reduce construction time and cost, as suggested by [57]. We imposed a maximum column diameter of 15 m [54]. We additionally constrained column height to a maximum of 20 m, the tallest observed in reported studies [46,58]. To achieve a proper liquid and gas distribution, the pressure drop across a single column is limited to a maximum of 0.002 bar and the liquid level to flooding is maintained below 70% [59]. If the flooding factor would exceed 70% at a diameter of 15 m, an additional column is introduced. Several studies, e.g., [54,60], suggest 20 stages in each absorption column and 10 in each stripper column, and we confirmed these values are appropriate for maximizing absorption and desorption in our capture block designs.

We chose to design columns with stainless steel packing, based on [60,61], and calculated performance of the packing section using the *Aspen* + rate-based model [60,62], which was validated using experimental data [63]. The goal of our modelling of the column was to estimate a purchased equipment cost (PEC), for which we require an estimate of column volume. For relatively large columns operating near atmospheric pressure, as in our designs, the PEC per unit volume ($/m^3) is dominated by cost of the packing [64]. Therefore, we chose to estimate column PEC (in Section 3.3) based on the volume of our modeled packing section. Given uncertainties, this may higher or lower than a cost estimate based on more detailed engineering.

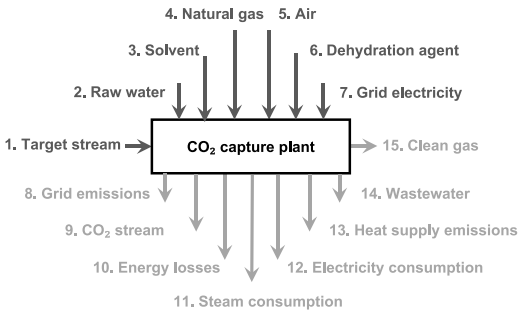
The **CO_2 compression** block is modeled with six centrifugal stages to bring the captured CO_2 to 86 barg, followed by a pump to boost pressure to 110 barg [35]. Pressure ratio per stage and isentropic and mechanical efficiencies of 2, 80%, and 90%, are assumed, respectively, based on [65]. The **CO_2 dehydration** block, which is placed between the last centrifugal compressor stage and the pump, reduces the water content from 25,000 ppm to <300 ppm to meet guidelines [66] for CO_2 transport in carbon steel pipes and injection into saline reservoirs for storage. We adopt a triethylene glycol (TEG) dehydration system, as used in the Shell Quest CCS project [35]. The equipment arrangement and specifications follow those of Bahadori et al. [67], and the required amount of TEG is based on modelling by Kemper et al. [68]. See equipment configuration details in SI.4.

The power-to-heat output ratio of the **cogeneration** block is set to exactly match the parasitic energy needs of the capture facility. Power-to-heat ratios (MW_e/MW_{th}) vary by cogeneration technology [69]. Considering that the power-to-heat ratios in our plant designs are approximately 0.2, we utilize boiler/back-pressure steam turbines with design inlet steam pressure of 130 barg and back-pressure of 18 barg. CO_2 emissions from a cogeneration facility are a function of the fuel type it uses [70] and whether or not flue gas CO_2 is captured. We assume natural gas is the boiler fuel, but future boiler modifications could offer compatibility with lower-carbon fuels [71]. In most of our plant designs, we assume that the cogeneration flue gas CO_2 is not captured, but we also design and evaluate one case where flue gas CO_2 is captured along with the target stream CO_2. Equipment configurations and design specifications for our cogeneration plant are as described by Jechura [72] and listed in the SI.6.

Our **cooling water system** utilizes a closed-cycle, which is estimated by Amin et al. [73] to have lower overall costs then an open-cycle cooling system (e.g., pond, river). Cooling water available at a design temperature of 15 °C is heated to 25 °C and cooled back down via a crossflow cooling tower [74,75]. The required water flow rate is calculated using a design specification in *Aspen* + and reported in SI.12.

3.2. Performance simulation results

Overall mass, carbon, and energy balances for facilities capturing 1 Mt/y of target-stream CO_2 are shown in Fig. 2. The figure includes results for target-stream CO_2 concentrations of 5%, 10%, and 15% when



Parameters	Cogeneration emissions are not captured				are captured
CO_2 concentration (mol%)	5%	10%	15%	94%	5%
Mass balance (tonnes per hour)					
1. Target stream	1,768	907	621	135	1,768
2. Makeup water	103	105	94	-	174
3. Makeup solvent	0.30	0.24	0.18	-	0.52
4. Natural gas	16	16	16	0.08	24
5. Air	447	407	404	2	606
6. Dehydration agent	0.01	0.01	0.01	0.01	0.01
7. Grid electricity	-	-	-	-	-
8. Grid emissions	-	-	-	0.01	-
9. CO_2 stream	131	131	131	131	194
10. Energy losses	-	-	-	-	-
11. Steam consumption	-	-	-	-	-
12. Electricity consumption	-	-	-	-	-
13. Heat supply emissions	463	423	420	2	-
14. Waste water	52	51	51	3	91
15. Clean gas	1,689	830	533	-	2,288
Carbon balance (tonnes of carbon per hour)					
1. Target stream	38	38	38	38	38
2. Makeup water	-	-	-	-	-
3. Makeup solvent	0.06	0.05	0.04	-	0.10
4. Natural gas	12	12	12	0.06	18
5. Air	-	-	-	-	-
6. Dehydration agent	0.001	0.001	0.001	0.001	0.001
7. Grid electricity	-	-	-	-	-
8. Grid emissions	-	-	-	-	-
9. CO_2 stream	36	36	36	36	53
10. Energy losses	-	-	-	-	-
11. Steam consumption	-	-	-	-	-
12. Electricity consumption	-	-	-	-	-
13. Heat supply emissions	12	12	12	0.06	-
14. Waste water	0.11	0.11	0.11	2	0.19
15. Clean gas	2	2	2	-	3
Energy balance (Megawatt)					
1. Target stream	-	-	-	-	-
2. Makeup water	-	-	-	-	-
3. Makeup solvent	-	-	-	-	-
4. Natural gas	247	242	243	1	366
5. Air	-	-	-	-	-
6. Dehydration agent	-	-	-	-	-
7. Grid electricity	-	-	-	18	-
8. Grid emissions	-	-	-	-	-
9. CO_2 stream	-	-	-	-	-
10. Energy losses	92	90	90	0.23	139
11. Steam consumption	132	130	130	1	195
12. Electricity consumption	23	22	22	18	32
13. Heat supply emissions	-	-	-	-	-
14. Waste water	-	-	-	-	-
15. Clean gas	-	-	-	-	-

Fig. 2. Mass, carbon, and energy balances for 1 $MtCO_2/y$ CO_2 capture facilities with CO_2 input concentrations of (a) 5 mol%, (b) 10 mol%, (c) 15 mol%, and (d) 94 mol%. Heat supply emissions are from cogeneration plant except for 94% case where it comes from a boiler.

cogeneration flue gas CO_2 is not captured and the case for 94 mol% target-stream concentration. It also includes the case for 5% target-stream concentration when cogeneration flue gas CO_2 is captured. (The total annual CO_2 captured in that case is 1.5 $MtCO_2/y$). Detailed *Aspen* + results for the 22 simulations with target streams of 15 mol% or less CO_2 concentration are provided in SI.12. Detailed results for the 94 mol% target stream concentrations are not shown separately in the SI, because the compression, dehydration and cooling-water blocks are identical in size and performance to those in cases with the same annual CO_2 capture rate but lower target-stream CO_2 concentrations (and no capture of cogeneration flue gas CO_2).

Carbon and energy balances for the three cases with the lowest-CO_2-concentration target streams and no capture of cogeneration flue gas CO_2 in Fig. 2 are essentially the same, because these are primarily a function of the amount of CO_2 captured. Mass flows of the target and clean gas streams vary, of course, due to the differing concentrations of CO_2 in the target stream. Makeup solvent requirements decrease with increasing CO_2 concentration as a consequence of less circulating solvent mass flow (calculated using Eq. 3) is needed for high CO_2 concentration, which is caused by the increase of rich amine solution loading at higher CO_2 concentration. Comparing the two cases in Fig. 2 with 5 mol% target stream concentration, one without and one with capture of cogeneration flue gas CO_2, we see that capturing the cogeneration CO_2 increases the load on the capture system by nearly 50%, with associated increases in use of electricity, makeup water, makeup solvent and natural gas, as well as waste water.

Our simulation results are consistent with other literature. We found the parasitic heat required for each capture system for low-CO_2 concentration target streams is around 3.6 GJ/tCO_2. Luo et al. found a value of 3.76 GJ/tCO_2 for a similar MEA-based CO_2 capture system [59]. Additionally, the steam demand is slightly higher for the 5 mol% CO_2 case than for 10 mol% or 15 mol%, as expected based on [76]. The overall HHV energy efficiency of the cogeneration block is 93% (84% steam, 9% electricity). Cogeneration systems utilizing backpressure-steam-turbines typically achieve thermal efficiencies between 70% and 85% and electric efficiencies between 5% and 10% [69].

Equipment capacities for each process simulation are given in SI.7 and provide the basis for estimating facility capital costs, as described in the next section. Equipment capacities for compression, dehydration and cooling-water blocks for the 94 mol% target stream concentrations are identical to those in cases with the same annual CO_2 capture rate but lower target-stream CO_2 concentrations (and no capture of cogeneration flue gas CO_2).

3.3. Capital cost estimates

We estimate the total installed capital cost, $C_{CAP,}$ for each of our simulated plants using Eq. 6,

$$C_{CAP} = TPEC \times LF \times SF \times JF \times RF \tag{6}$$

where *TPEC* is the estimated total purchased equipment cost determined as the sum of purchased equipment costs (PEC_i) for the capture, compression, dehydration, cogeneration, and cooling blocks:

$$TPEC = PEC_{CAP} + PEC_{COM} + PEC_{DEH} + PEC_{COG} + PEC_{COL} \tag{7}$$

Purchased equipment costs are estimated (1) using sizes of individual components within each block from our *Aspen* + simulations, and (2) using *CapCOST*, an open-source database for estimating equipment costs based on design specifications and capacities [64], except for pumps and compressors, for which we used [77], steam turbines, for which we used [78], and cooling water systems, for which we used [74,79]. Costs are expressed in 2020 US dollars escalated from source data using the Chemical Engineering Plant Cost Index (CEPCI).

The multipliers in Eq. 6 are the Lang factor (*LF*) [80], which accounts for balance of plant equipment and all plant construction costs, a process-contingency factor (*SF*), a project contingency factor (*JF*), and a retrofit factor (*RF*). These are discussed further later in this section.

Detailed *TPEC* estimates for all target streams (5, 10, 15, and 94 mol % CO_2) are available in SI.8. For illustration, Fig. 3 summarizes our *TPEC* estimates for 5 mol% CO_2 target streams when cogeneration CO_2 emissions are not captured. At all scales, the dehydration block contributes only a minor amount to *TPEC*. At the smallest scale, the other four blocks each contribute 20–30%. At any scale, the share of *TPEC* contributed by the cogeneration block remains at roughly 25%. Shares contributed by compression and cooling water systems decrease with increasing scale, while the capture block grows from 30% to 60% of *TPEC*.

As noted earlier, the plant design with 1.15 $MtCO_2/y$ capture capacity was also modeled with cogeneration CO_2 emissions captured. For that design, the capture capacity increases to 1.70 $MtCO_2/y$, and the *TPEC* increases by 41% to $114 million. PEC for the capture and cogeneration blocks grow by 48% and 38%, respectively. See SI.8 for additional details.

In the capture block for all designs, the main cost contributions come from absorption columns, the total volume of which is significantly larger than the stripper volume (Fig. 4). Absorber volume increases non-linearly with decreasing CO_2 concentration to accommodate the non-linearly increasing target-stream mass flow. For a given rate of CO_2 capture, stripper volume remains roughly constant regardless of target stream CO_2 concentration because the CO_2 removal duty is the same. Given the column height and diameter design constraints noted earlier, the capture block employs varying numbers (and sizes) of absorber and stripper columns across different cases to realize the column volumes required. See the SI.7 for details.

Table 2 lists the values used for each of the *TPEC* multipliers in Eq. 6 to determine total installed capture plant capital cost, C_{CAP}. Values are given for three different levels of assumed technology maturity. Early-mover projects might typically relate to the second and third projects in a given geographic region (or the 10th to 20th projects world-wide); N^{th}-of-a-kind estimates would apply to those beyond the early mover projects [81]. Of course, all such projections will depend very much on whether new projects have the access and capacity to learn from previous projects. A process contingency of 50% for a first-of-a-kind (FOAK) project implies a process design based on limited data or bench scale data, dropping to zero for an Nth-of-a-kind (NOAK) commercial plant [82]. Project contingencies between 30% and 50% of *TPEC* reflect the scoping-study nature of our plant designs in contrast to smaller values that would be appropriate after detailed engineering work [82]. Retrofit factors reported in the literature vary from 5% [83] to 20% of *TPEC* [84,85]. We use 5%, assuming there would be only minor modifications needed to existing facilities, since capture streams otherwise would be vented directly to the atmosphere.

Calculated NOAK installed capital costs per unit of capture capacity for 5 mol% CO_2 target streams (without capture of cogeneration CO_2 emissions) as a function of capacity are shown as dots in Fig. 5, together with the corresponding regression curve fit. Regression curves for other target stream CO_2 concentrations (without capture of cogeneration CO_2 emissions) are also shown. All curves have the following form:

$$C_{CAP} = C_0 \times \dot{M}_{CO_2}{}^{-b} \tag{8}$$

where C_{CAP} is total installed capital cost (2020$ per tCO_2/yr), $\dot{M}_{CO_2}$ is the installed capture capacity (tCO_2/yr), and parameters C_0 and b depend on target stream CO_2 concentration, and assumed level of commercial maturity (Table 3).

By way of validating our cost model, we find that it gives results that are largely consistent with published cost estimates for three actual CCS retrofit projects that included dedicated cogeneration plants whose emissions are not captured: Boundary Dam [34,49], Quest [36], and Petra Nova [37] (Table 4). These plants can reasonably be considered first-of-a-kind projects. Our cost estimate for Boundary Dam as a FOAK project is 3% lower than the published estimate, but our estimates for

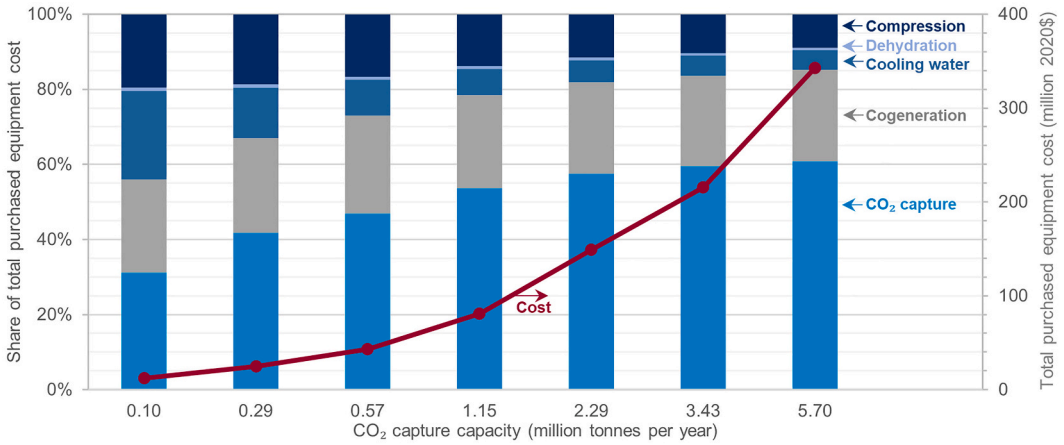


Fig. 3. Total purchased equipment costs (red circles, right axis) and breakdown by process block (bars, left axis) for 5 mol% CO_2 target streams and a range of annual CO_2 capture capacities when cogeneration CO_2 emissions are not captured. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

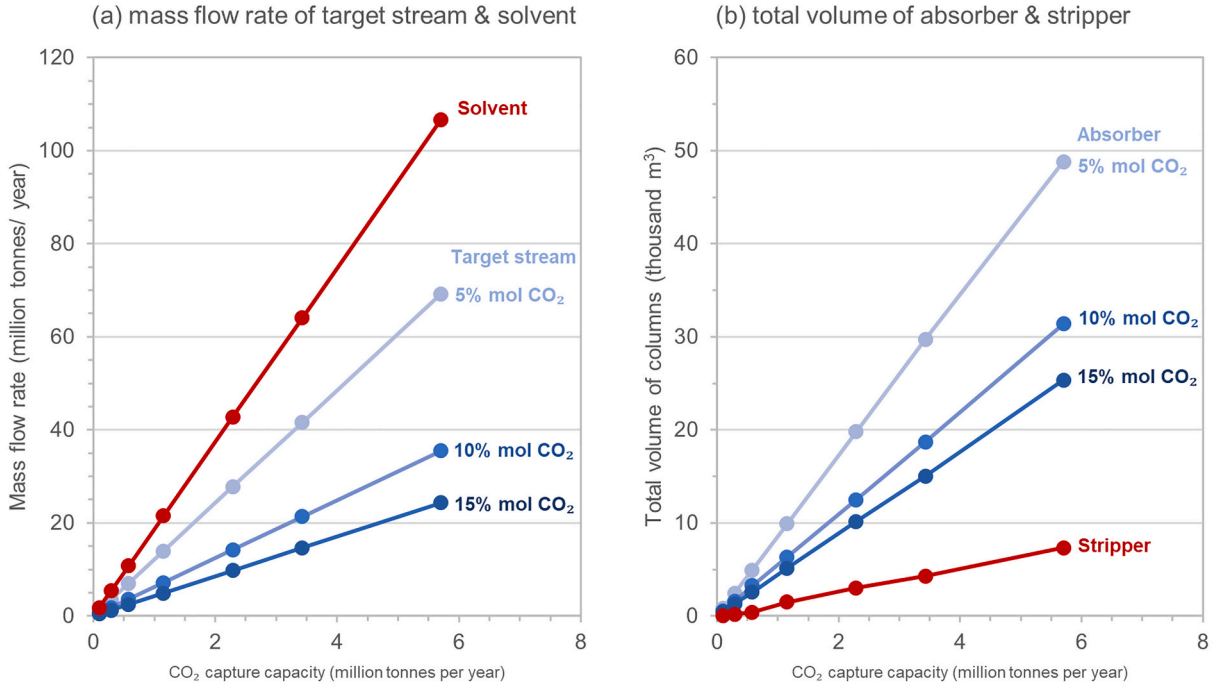


Fig. 4. (a) Mass flow rate of target streams and lean solvent and (b) absorber and stripper volumes, all as functions of annual CO_2 capture capacity for designs without capture of cogeneration CO_2 emissions. Solvent flow and stripper volume are shown for only for 15 mol% CO_2 target streams, but they are nearly the same for 5 mol% and 10 mol% target streams – see SI.7 and SI.12.

Table 2
Cost factors for different technological maturity levels of CO_2 capture systems.

Parameters	Cost factors			Ref.
	First-of-a-kind	Early movers	N^{th}-of-a-kind	
Lang factor, *LF*	6.21	6.21	6.21	[80]
Process contingency, *SF*	1.50	1.30	1.00	[82,86,87]
Project contingency, *JF*	1.50	1.40	1.30	[22,82,87]
Retrofit factor, *RF*	1.05	1.05	1.05	[84,85,87]
Total cost multiplier	14.67	11.87	8.48	

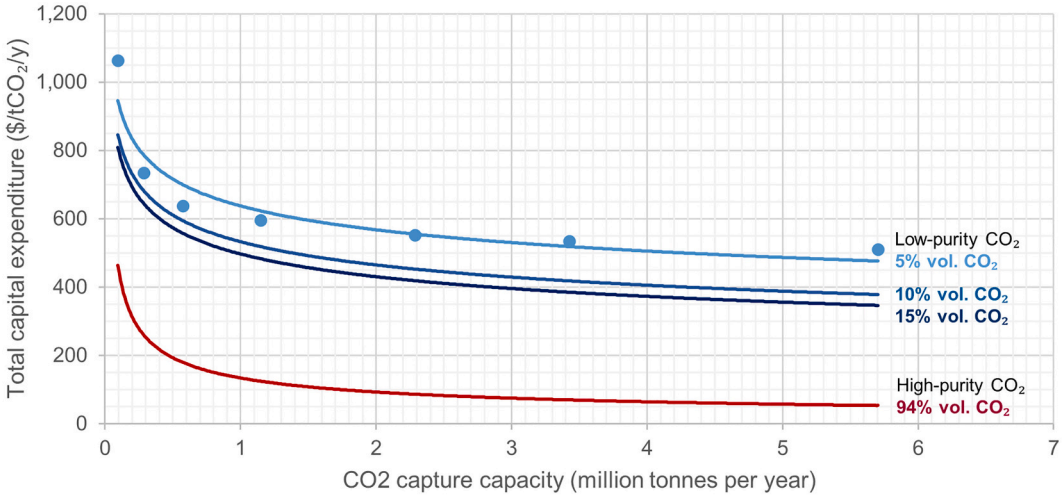


Fig. 5. Regression curve fits for estimating total installed capital cost of CO_2 capture for N^{th}-of-a-kind plants.

Table 3
Parameter values for Eq. 6.

% mol CO_2	5%	10%	15%	94%
N^{th}-of-a-kind (NOAK)				
C_0	637	532	496	133
b	0.167	0.197	0.207	0.528
R^2	0.91	0.92	0.92	0.99
Early mover (EM)				
C_0	892	745	695	133
b	0.167	0.197	0.207	0.528
R^2	0.91	0.92	0.92	0.99
First-of-a-kind (FOAK)				
C_0	1103	922	860	133
b	0.167	0.197	0.207	0.528
R^2	0.91	0.92	0.92	0.99

Quest and Petra Nova as FOAK projects are over 15% higher, through still with in the uncertainty bounds of such estimates. The lower published estimates for these two projects may stem from lower design capture rates for those plants (80–90%), the use of some waste heat that reduced the needed capacity of the dedicated cogeneration block providing parasitic energy needs, and/or the exclusion of certain ancillary costs from the published estimates. Early-mover projects might typically relate to the second and third projects in a region (or between 10th and 20th projects world-wide), and projects beyond those might be categorized as N^{th}-of-a-kind [81]. In the real world, all such projections will depend very much on whether new projects have the access and capacity to learn from previous projects.

3.4. Levelized cost of CO_2 capture

The capital cost estimates developed above, together with fixed and variable operating costs (discussed below) provide inputs for estimating total levelized costs of CO_2 capture (*LCC*), as follows

$$LCC = \frac{(CRF \times C_{CAP}) + OPEX_{CAP,FIX} + OPEX_{CAP,VAR}}{0.9 \times \dot{M}_{CO_2,CAP}} \quad (9)$$

where C_{CAP} is total installed capital cost for the capture plant, and *CRF* is the annual capital recovery factor,

$$CRF = \frac{\left(i \cdot (1+i)^T\right)}{\left((1+i)^T - 1\right)} \quad (10)$$

with weighted average cost of capital, *i*, and asset economic life, *T* (in years). $\dot{M}_{CO_2,CAP}$ is as defined earlier (annual amount of CO_2 captured if the plant is operating at full capacity all year), and we assume a 90% annual plant capacity factor. (The actual annual CO_2 capture rates corresponding to the seven values of $\dot{M}_{CO_2,CAP}$ given earlier are 0.09, 0.26, 0.52, 1.03, 2.06, 3.08, and 5.13 $MtCO_2$/y.) Finally, $OPEX_{CAP,FIX}$ and $OPEX_{CAP,VAR}$ are annual fixed and variable operating costs:

$$OPEX_{CAP,FIX} = C_{LAB} + C_{A\&O} + C_{O\&M} + C_{ITF} \quad (11)$$

$$OPEX_{CAP,VAR} = C_{MEA} + C_{H2O} + C_{ELC} + C_{NAT} \quad (12)$$

Fixed operating cost includes direct labor costs (C_{LAB}), administrative and general overhead costs ($C_{A\&O}$), annual operating and

Table 4
Comparison of CO_2 capture plant cost estimates from this work (Eq. 8) with published cost estimates (adjusted to 2020$ using CEPCI indices) for actual CO_2 capture projects: Boundary Dam [34,49], Quest [36], and Petra Nova [37,50].

Project name		Boundary Dam	Quest	Petra Nova[1]
Location		Saskatchewan, Canada	Alberta, Canada	Texas, USA
Operation year		2014	2015	2017
Designed capacity of CO_2 capture	(million tonnes / year)	1	1	1.6
CO_2 concentration in target stream	(mol%)	11.4	18	14
Published total capital expenditure	(million 2020$)	962[2]	663	699
Estimated total capital expenditure				
N^{th}-of-a-kind[3]	(million 2020$)	537	480	464
Early mover[3]	(million 2020$)	752	672	650
First-of-a-kind[3]	(million 2020$)	931	832	804

Notes
[1)]Operation was suspended in 2020.
[2)]Excluding coal plant refurbishment costs.
[3)]By linear interpolation (or extrapolation) from modeled concentration levels.

Table 5
Financial parameters.

Economic parameters	Units	Value	Ref.
System			
Weighted average cost of capital, i	%	10	[58,75]
Economic lifetime, T	years	30	[75,89]
Capital recovery factor, CRF	1/year	0.106	From Eq. 10
Capacity factor, λ	%	90	[58,82]
Fixed costs			
Administrative and general overhead cost, $C_{A\&O}$	$	30% of C_{LAB}	[58,82]
Annual operating and maintenance cost, $C_{O\&M}$	$	1.5% of C_{CAP}	[58,82]
Insurance, local taxes and fees, C_{ITF}	$	1% of C_{CAP}	[58,82]
Variable costs			
Natural gas price, c_{NAT}	$/GJ$_{HHV}$	3[a]	[90]
Electricity price, c_{ELC}	$/MWh	49[b]	[91]
Water price, c_{H2O}	$/m^3	0.01	[92]
Monoethanolamine (MEA), c_{MEA}	$/tonne	2880	[92]

(a) Average natural gas price to industrial users in Louisiana in 2020 [90].
(b) Average electricity price to industrial customers in Louisiana in 2020 [91].

maintenance costs ($C_{O\&M}$), and insurance, local taxes and fees (C_{ITF}). Direct labor costs, as detailed in SI.10, assume four working shifts to achieve 90% capacity factor [64], two operators per shift for plants capturing ≤0.4 MtCO₂/y [58] and between one and three additional operators per shift for larger plants. An annual average salary and benefits of $110,000 per operator is based on U.S. Bureau of Labor Statistics data for industrial engineering technologists and technicians [88]. Assumed values for the other components of fixed operating costs in Eq. 11 are as shown in Table 5.

Variable operating costs (Eq. 12), include costs for solvent (C_{MEA}), water (C_{H2O}), electricity (C_{ELC}), and natural gas (C_{NAT}). Prices assumed for each of these are given in Table 5.

Our calculated levelized costs of CO₂ capture (LCC) for all 28 simulated process designs when cogeneration CO₂ emissions are not captured are shown in Fig. 6, assuming NOAK facilities. Scale economies are evident in the capital and fixed operating cost components, regardless of the CO₂ concentration of the target stream. Also, it is noteworthy that costs associated with the cogeneration block, including capital and natural gas fuel purchases, are significant. These would be reduced if a capture facility were able to utilize waste heat from the host industrial facility it serves. Given the capital-intensity of the simulated capture plants, especially for capture rates of 1 MtCO₂/year or less, it is unsurprising that our sensitivity analysis across a number of different parameters (Fig. 7) shows uncertainties in capital-related parameter values to have the most significant impact on levelized capture cost: weighted average cost of capital, total installed capital cost, and annual capacity factor have larger impacts than the same percentage change in a variety of other techno-economic parameter values.

To estimate capture costs for industrial target streams having arbitrary CO₂ concentrations and mass flows (without capture of cogeneration CO₂ emissions), regression curves were fit to the LCC values shown in Fig. 6. The curves have the form

$$LCC = C_0 \times \left(0.9 \times \dot{M}_{CO_2}\right)^{-b} \tag{13}$$

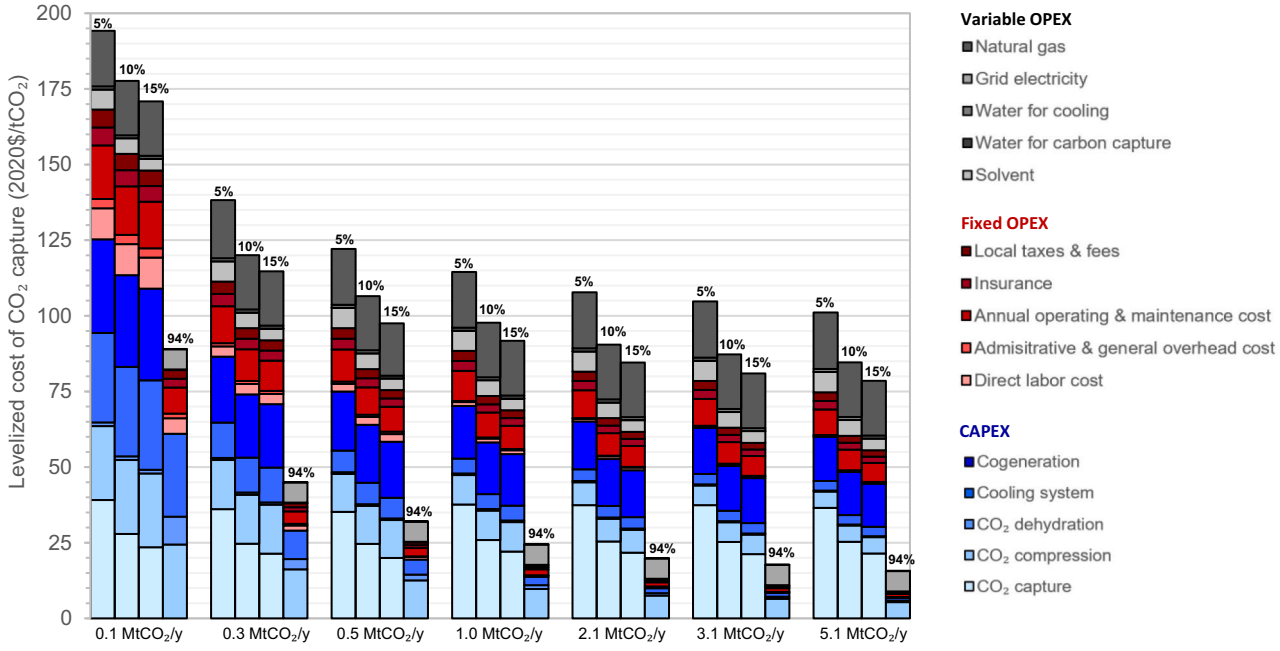


Fig. 6. Levelized CO₂ capture cost estimates for 28 simulated capture plants, assuming Nth-of-a-kind plants and no capture of cogeneration CO_2 emissions. Target-stream CO₂ concentration (mol%) is shown above each bar. The annual capture rate (x-axis) is for capture of 95% of the target-stream CO₂ and 90% annual plant capacity factor.

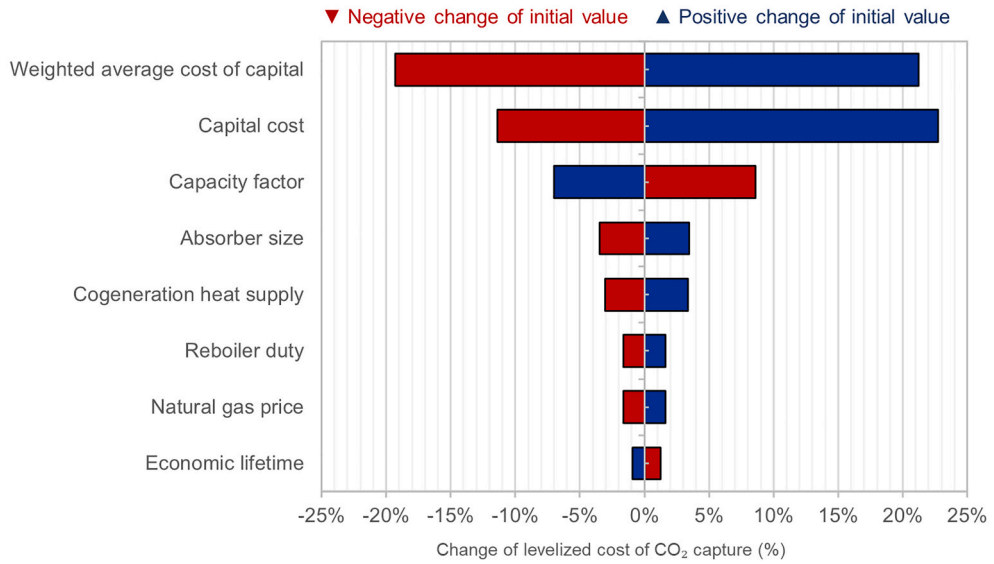


Fig. 7. Sensitivity of levelized CO_2 capture cost to techno-economic input parameter values for an Nth-of-a-kind plant with target stream containing 5 mol% CO_2 and annual capture rate of 1 $MtCO_2$/yr. All parameters are changed to ±10% of their initial values except for weighted average cost of capital (±40%) and capital cost (−15% & +30%) to suitably reflect the nature of uncertainties for these parameters.

Table 6
Coefficient and exponent values for Eq. 13.

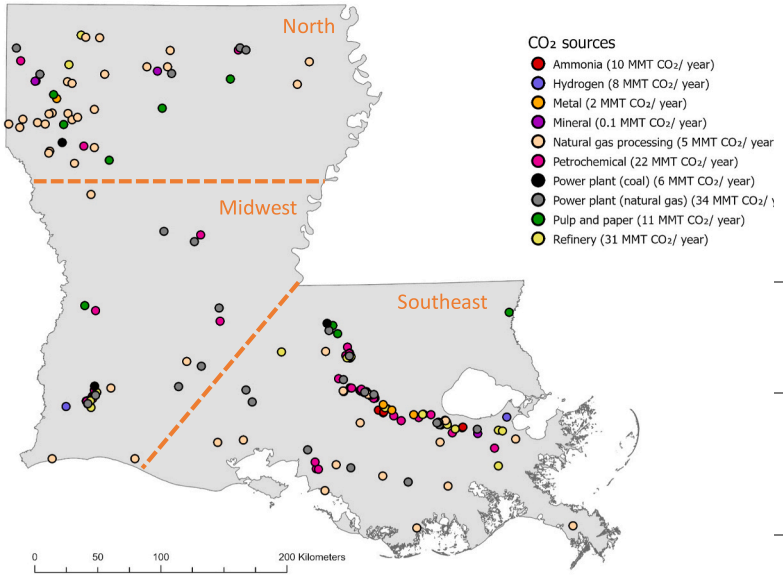
% mol CO_2	5%	10%	15%	94%
N^{th}-of-a-kind (NOAK)				
C_0	123	105	99	27
b	0.146	0.167	0.175	0.415
R^2	0.90	0.91	0.90	0.97
Early mover (EM)				
C_0	159	136	127	27
B	0.151	0.174	0.182	0.415
R^2	0.90	0.91	0.91	0.97
First-of-a-kind (FOAK)				
C_0	189	161	151	27
b	0.154	0.177	0.186	0.415
R^2	0.90	0.91	0.91	0.97

where LCC is the levelized cost of capture (2020$/$tCO_2$), $\dot{M}_{CO_2}$ is the annual CO_2 capture rate (t/y) and values of C_o and b are as in Table 6.

The levelized costs estimated using Eq. 13 and Table 6 would be slightly lower if facilities were to include capture of cogeneration CO_2 emissions: the absolute capital cost would be higher (as illustrated in Section 3.3), but because of PEC scale economies (reflected in Fig. 6) the levelized cost per tonne of CO_2 captured would be lower. For example, in the case of a 5 mol% CO_2 target stream, for which the annual capture capacity would increase from 1.15 $MtCO_2$/y without cogeneration CO_2 capture to 1.70 $MtCO_2$/y with cogeneration CO_2 capture, the levelized cost per tonne of CO_2 captured would decrease by 3%, as detailed in SI.9.

4. Louisiana case study of CTS system designs

We illustrate application of our capture cost model with a case study



Region	Number of facilities	CO_2 emissions ($MtCO_2$/y)		
		Total	Per facility	
			Average	Median
Midwest	34	31	0.92	0.07
North	49	15	0.31	0.37
Southeast	107	83	0.75	0.31
Louisiana	190	130	0.67	0.19

Fig. 8. Industrial facilities in Louisiana reporting emissions of 25,000 t/y or more of CO_2 in 2019 [96]. Aggregated total emissions by type of facility are shown in the legend. The Southeast region accounts for 64% of total industrial emissions and the largest number of individual emitting facilities. SI.2 provides details for each facility shown here.

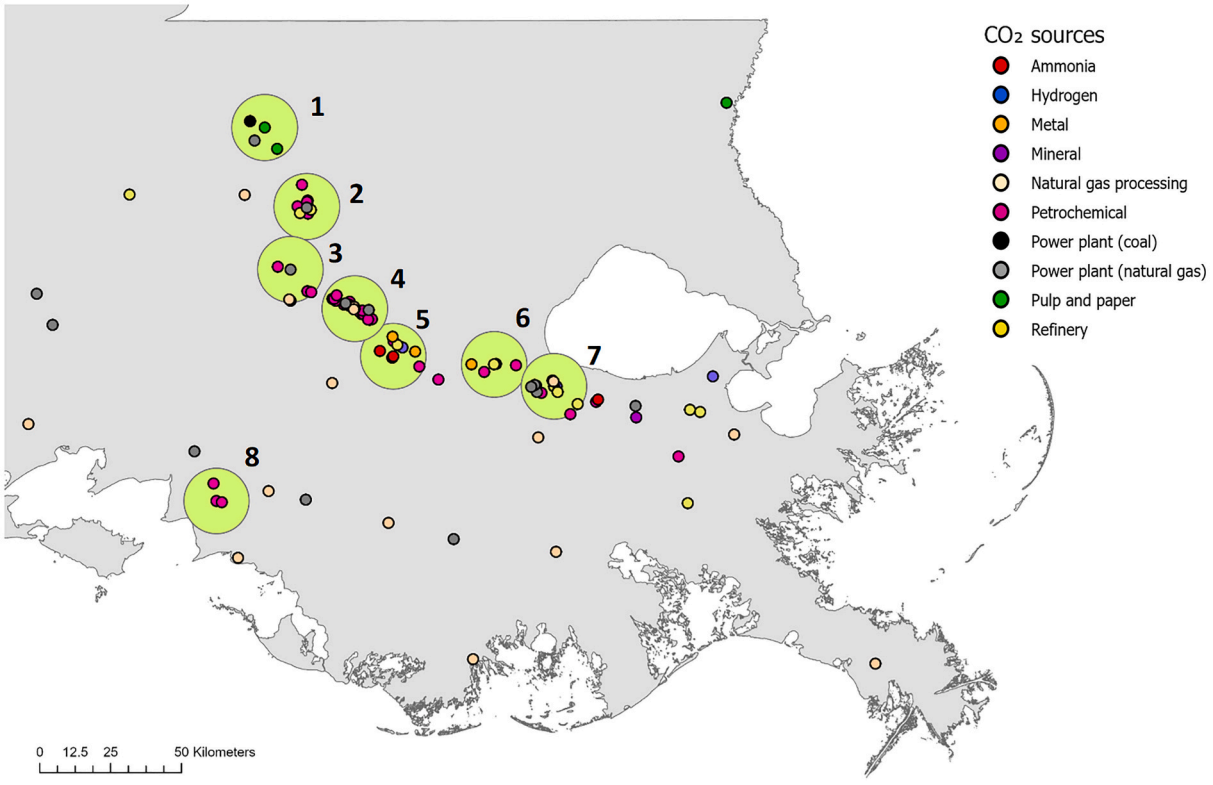


Fig. 9. Industrial clusters in Southeast Louisiana.

analysis of alternative cost-optimized configurations for capture, transport, and storage (CTS) of CO_2 from clusters of industrial facilities in the U.S. state of Louisiana, which has among the best underground CO_2 storage resources in the U.S. [93], along with a well-established CO_2 pipeline transport industry. It is also one of the most industry-dense US states, with 190 industrial facilities that emitted 130 $MtCO_2$ of CO_2 in 2019, or nearly three-quarters of the state's total emissions [94] (Fig. 8). In 2020, Louisiana's governor set targets to cut the state's greenhouse gas emissions 40–50% by 2030, relative to 2005, and to net-zero by 2050 [95]. CO_2 capture and storage is envisioned to be one element of the strategy for achieving these goals [95].

4.1. Case-study industrial facilities and CO_2 capture costs

Nearly two-thirds of Louisiana's industrial emissions are from facilities in the Southeast part of the state (Fig. 8), which may be typical of heavily industrialized regions found worldwide. We focus our case study analysis on this region, and examine in particular how relatively smaller facilities (with emissions $\leq$1 $MtCO_2/y$), which incur higher costs for capture (Fig. 6) and account for 25% of Louisiana's industrial emissions (78% of Louisiana's industrial facilities), might benefit through sharing of infrastructure with other facilities, potentially including shared capture plants in addition to shared CO_2 transport and storage infrastructure. Since facilities in close physical proximity to each other may be able to benefit more from shared infrastructure than more-separated facilities, we first identify clustered groups of facilities using a geoprocessing algorithm in *ArcGIS*^{PRO} that identifies circular areas of radius 10 km that each include 3 or more unique facilities. The 10 km radius was chosen considering that under a potential scenario in which facilities send their emission streams to a shared capture plant, a prior study [45] suggests transport of flue gases up to 14 km would be feasible.

Fig. 9 shows the result of the geoprocessing analysis. Eight clusters encompass 77 facilities (of the total 111 in the region) responsible for 67 $MtCO_2/y$ of emissions, or 90% of industrial CO_2 emissions in the region. Of these 77 facilities, 22 are small (emissions <0.1 $MtCO_2/y$), 35 are medium-sized (emissions from 0.1 $MtCO_2/y$ to 1 $MtCO_2/y$), and 20 are large (emissions >1 $MtCO_2/y$), as shown in Table 7. The SI.2 lists CO_2 emissions in 2019 for each facility and other facility details. For evaluating trade-offs across different designs for shared CO_2 capture, transport, and storage infrastructure we selected the cluster with the highest

Table 7
Number of facilities, CO_2 emissions, and CO_2 density for each cluster identified in Fig. 9.

Cluster	Number of facilities				CO_2 emissions, $\dot{M}_{CO_2,z}$ ($MtCO_2/y$)				CO_2 emissions density ($ktCO_2/km^2$)			
	All	Small	Med	Large	All	Small	Medium	Large	All	Small	Medium	Large
1	4	1	2	1	4.1	0.09	1.1	2.91	13.1	0.28	3.5	9.27
2	12	7	3	2	10.6	0.32	1.9	8.38	33.7	1.01	6.04	26.7
3	7	0	4	3	7.17	0	1.21	5.96	22.8	0	3.85	19
4	22	9	10	3	8.12	0.42	4.24	3.45	25.9	1.35	13.51	11
5	9	2	5	2	13.4	0.12	2.98	10.3	42.7	0.39	9.49	32.8
6	5	1	3	1	5.76	0.04	1.78	3.94	18.3	0.13	5.67	12.6
7	15	2	5	8	17.8	0.09	1.91	15.8	56.8	0.29	6.1	50.4
8	3	0	3	0	0.79	0	0.79	0	2.51	0	2.51	0

Note: Small sources ($\dot{M}_{CO_2,F} < 0.1$ $MtCO_2/y$), medium sources (0.1 $MtCO_2/y < \dot{M}_{CO_2,F} < 1$ $MtCO_2/y$), and large sources ($\dot{M}_{CO_2,F} > 1$ $MtCO_2/y$).

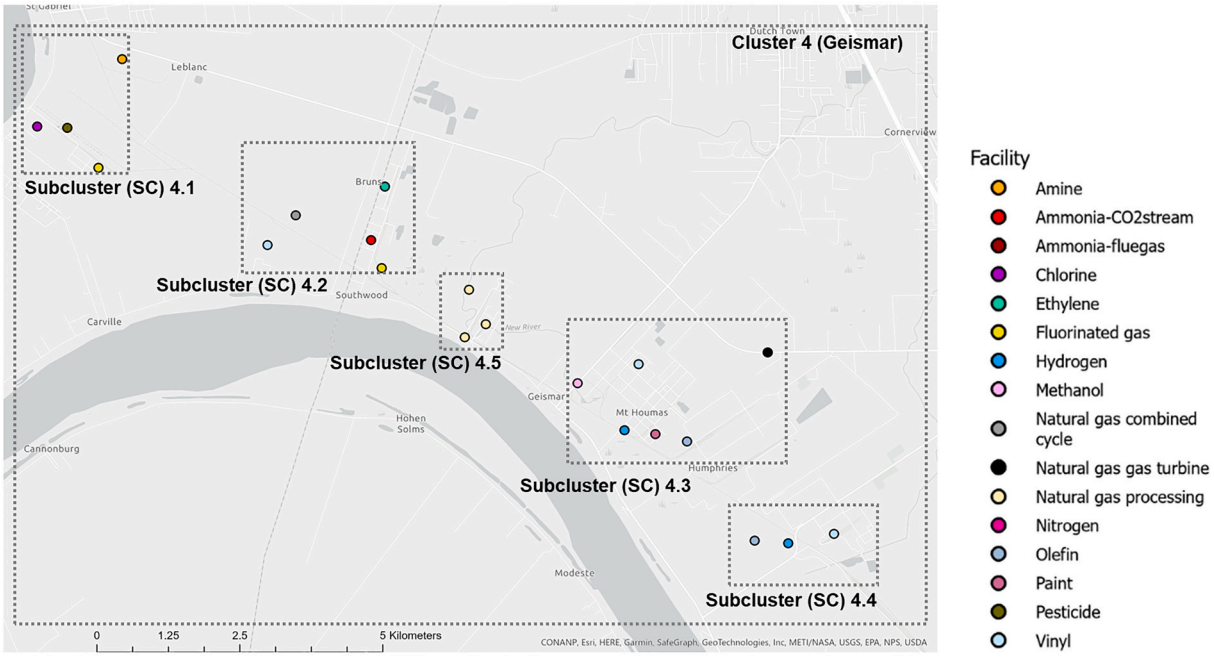


Fig. 10. Subclusters of CO_2 emitting facilities in cluster 4 (Geismar, Louisiana).

Table 8
List of clusters and associated facilities in Cluster 4 (Geismar).

Sub-cluster	Sector	Industrial output	CO_2 concentration		Capture rate**
			(mol%)	References	($MtCO_2$/y)
4.1	Petrochemical	Fluorinated gas	8	[97–102]	0.0338
	Petrochemical	Chlorine	8	[103–105]	0.0187
	Petrochemical	Pesticide	8	[106]	0.0511
	Petrochemical	Amine	8	[16,107]	0.0500
Subtotal-4.1 (weighted average)			8		0.1537
4.2	Petrochemical	Vinyl	8	[103–105]	0.3405
	Petrochemical	Fluorinated gas	8	[97–102]	0.1053
	Petrochemical	Ethylene	8	[24,108–110]	0.7347
	Power plant	Electricity (nat. Gas comb. cycle)	4	[23,75,111–114]	1.2670
	Ammonia*	Flue gas	6	[115–117]	0.4297
	Ammonia*	CO_2 stream	94	[115–117]	0.6025
Subtotal-4.2 (weighted average)			6		3.4797
4.3	Petrochemical	Nitrogen	8		0.3199
	Petrochemical	Olefin	8	[24,108–110]	0.7452
	Hydrogen	Hydrogen	13	[24,54,58,118,119]	0.5663
	Petrochemical	Methanol	10	[120–122]	0.7426
	Petrochemical	Paint	8	[123,124]	0.0605
	Petrochemical	Vinyl	8	[103–105]	0.2049
	Power plant	Electricity (nat. Gas gas turbine)	4	[23,75,111–114]	0.1483
Subtotal-4.3 (weighted average)			9		2.7876
4.4	Hydrogen	Hydrogen	13	[24,54,58,118,119]	0.0568
	Petrochemical	Vinyl	8	[103–105]	0.1238
	Petrochemical	Olefin	8	[24,108–110]	1.0101
Subtotal-4.4 (weighted average)			8		1.1907
4.5	Natural gas proc.	Natural gas	94	[28,51,89,125,126]	0.0052
	Natural gas proc.	Natural gas	94	[28,51,89,125,126]	0.0048
	Natural gas proc.	Natural gas	94	[28,51,89,125,126]	0.0030
Subtotal-4.5 (weighted average)			94		0.0130
Total Cluster 4 (excluding 94% CO_2 concentration streams)			8[1]		7.0093
Total Cluster 4 (including all streams)					7.6247

* The single ammonia-producing facility in the cluster has a high- and a low-CO_2 purity emission stream. We treat these as two separate target streams.
** Annual capture rate, assuming 90% capacity factor and no capture of cogeneration CO_2 emissions.

density of CO_2 emissions from small and medium facilities, Cluster 4 (Table 7), which is in the vicinity of Geismar, Louisiana.

Cluster 4 consists of five sub-clusters within each of which no facility is >2 km distant from any other facility (Fig. 10). Estimated CO_2 concentrations of target capture streams at each facility and estimated annual capture rates (assuming a 95% capture rate and a 90% plant capacity factor) are given in Table 8.

With the estimated CO_2 concentrations and flow rates, the levelized cost of CO_2 capture at each of the 22 facilities in Cluster 4 can be estimated using Eq. 13. Given the close proximity of facilities to each other in the case study region, it is also of interest to estimate the capture cost if multiple facilities were to share a capture plant and thereby benefit

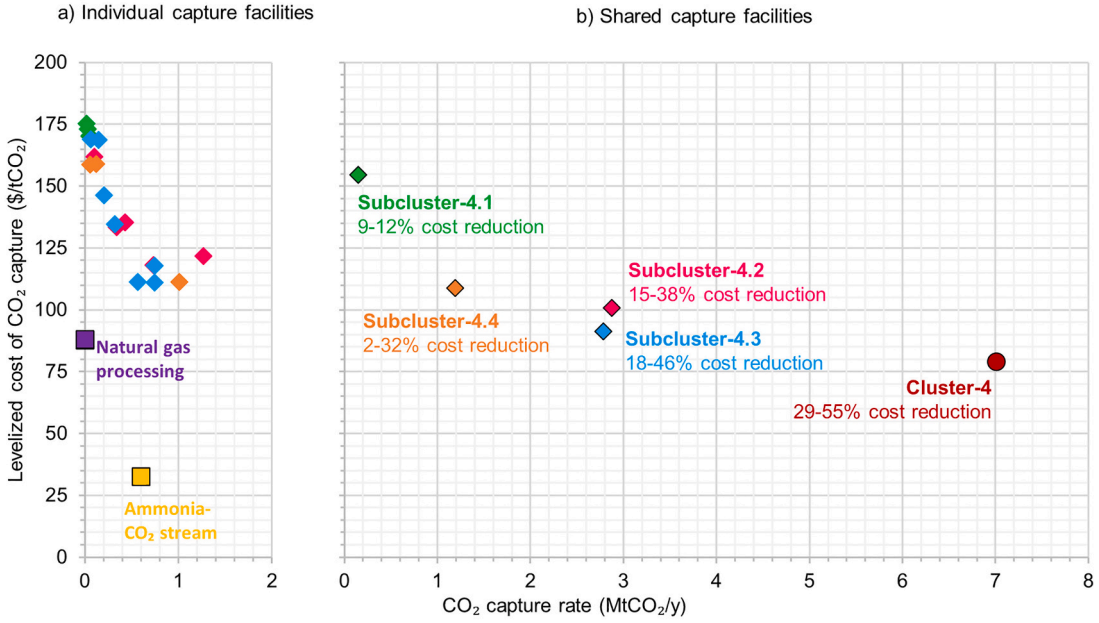


Fig. 11. The levelized cost of CO_2 capture ($/t$CO_2$) for capture plants (a) at individual facilities and (b) shared by facilities in a sub-cluster or shared by all facilities in the cluster. (Note: high-purity CO_2 emissions from natural gas processing and ammonia are not considered for the shared capture plants.)

from economies of scale. For example, each of the facilities in a single sub-cluster might share a capture plant or all facilities across the entire cluster might share a capture plant. For modelling shared CO_2 capture systems, we assume a target stream flow rate equal to the sum of flue gas flows from each facility and a CO_2 concentration equal to the weighted average of concentrations in flue gas from each facility. This, of course, would require transporting CO_2 from emitting facilities to the capture plant. We evaluate one approach for CO_2 transportation in detail in the next section.

Fig. 11 shows the levelized cost of capture for different levels of sharing. The squares are capture costs for individual high-CO_2 concentration target streams (e.g., ammonia or natural gas processing). The border-less diamonds are costs for individual capture plants at each of the 19 facilities with low-CO_2 concentration target streams. Diamonds of the same color are facilities in the same sub-cluster. The black-bordered diamonds are capture cost estimates when all facilities having low concentration target streams in a sub-cluster pool target streams and share a capture plant. Cost reductions for individual facilities range from 12% with the smallest shared capture plant for a sub-cluster to 38% with the largest sub-cluster capture plant. When all 19 facilities pool emissions and share a single capture plant (circle), the capture cost savings are as high as 55% relative to capture at an individual facility. Fig. 11 does not consider the costs of flue gas transportation that would be incurred when facilities share a capture plant, nor are the costs for transport and storage of captured CO_2 considered. These considerations are addressed next.

4.2. CO_2 transport and storage modelling

For Cluster 4, we consider alternative system designs (described in the next section) for capture, transport, and storage of the 7.6 $MtCO_2$/yr of CO_2 deemed to be capturable across all facilities when cogeneration CO_2 is not captured (Table 8). For each design, we consider as the CO_2 storage sites two underground formations studied by Dismukes et al. [94]. Bayou Sorrel, in Iberville Parish, is a depleted oil field approximately 32 km west of Cluster 4. It has an estimated storage capacity of 338 $MtCO_2$ and an average injectivity of 11 $MtCO_2$/y. Paradis, in St. Charles Parish, an EOR site about 67 km southeast of Cluster 4. Its estimated storage capacity is 30 $MtCO_2$ and average injectivity is 1 $MtCO_2$/y [94]. Average injection, storage, and monitoring costs for both sites are estimated to be 10 $/t$CO_2$ (in 2020$) [94].

For modelling of CO_2 pipeline capacities and routings from capture sites to the storage sites, we utilize *SimCCS*^{PRO}, a customized version of an open-source mixed-integer linear program for cost-optimizing the design of CO_2 pipeline networks [127]. The algorithm minimizes the total capture, transportation, and storage system cost for a given set of CO_2 sources and sinks and a targeted total amount of CO_2 capture and storage within a defined geographic region. Because pipeline costs depend on pipe diameters and lengths as well as routing topography [128], a pipeline cost model from [129] that encompasses pipeline materials, construction, right-of-way acquisition, and operating costs, is embedded in *SimCCS*^{PRO}, along with a "cost surface" [130] that adjusts the pipeline-model costs according to weights assigned to topographic features of the pipeline routes. The "cost surface" accounts, at 60–120 m resolution, for land cover type, land owner (public vs. private), land value, existing right-of-ways (natural gas pipelines, electricity transmission, railways, roadways), terrain slope, and rivers. *SimCCS*^{PRO} additionally assumes pumping stations are located every 80 km along a pipeline to maintain a CO_2 a pressure between 86 and 150 barg.

One additional form of CO_2 transport is also considered in some of our system designs: flue gases ducted from individual facilities to a shared CO_2 capture plant, an option suggested in some previous studies [32,45]. For routing of ducts, we use *SimCCS*^{PRO} to determine the cost-optimal routing assuming CO_2 pipelines connect the flue gas source to the shared CO_2 capture plant, and assume that the ducts follow the same routing. To estimate the cost for the ducting, we adapt the method and input values of Berghout et al. [45], where total installed capital cost is the sum of purchase cost for fans (C_B) and duct materials (C_S), plus construction (C_K). We use *CapCOST* [64] to estimate C_B based on the volumetric flow of flue gas and required pressure boost. An axial tube fan is assumed to be installed at the capture facility to increase flue gas pressure to 1.18 bara (see SI.11). C_S is calculated from:

$$C_S = \left(\frac{\pi \times D_{IN} \times L \times d \times \rho_S \times c_S}{0.95} + \pi \times D_{OUT} \times L \times c_C \right) \times 1.32 \qquad (14)$$

where circular ducts are assumed with inner diameter, D_{IN}, outer diameter, D_{OUT}, length, L, wall thickness, d, density of wall material, ρ_S, price per unit mass of the material, c_S, and price of wall coating, c_c. C_K is

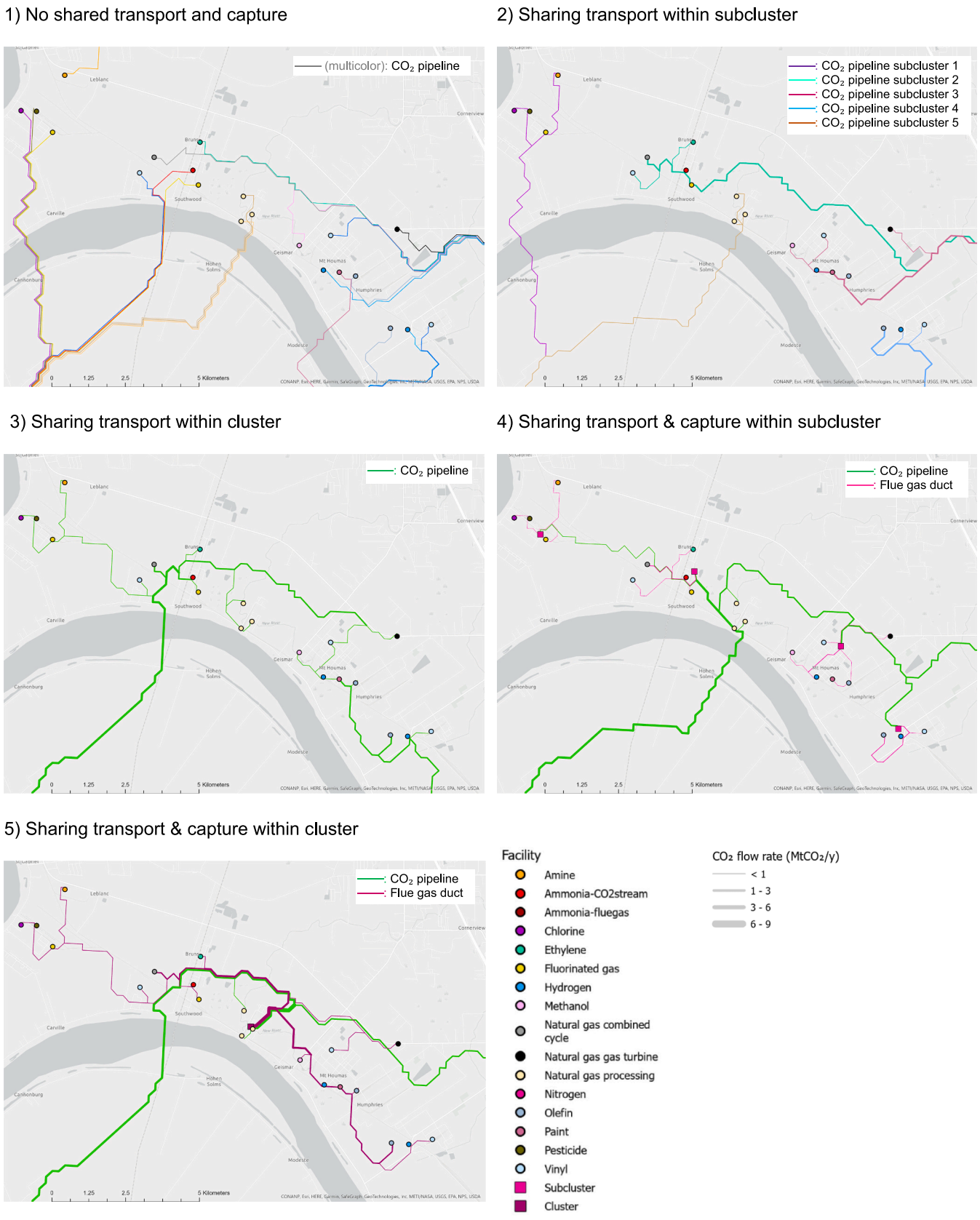


Fig. 12. Cost-optimal CO_2 pipeline and flue gas duct networks for five scenarios described in the text.

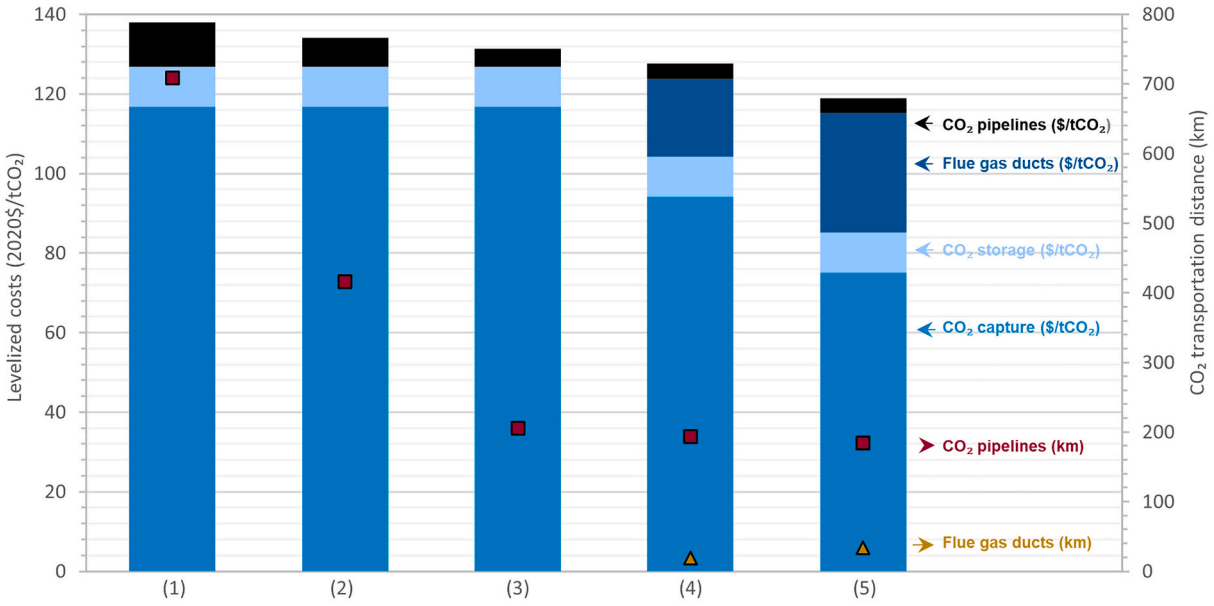


Fig. 13. Total levelized cost of CO_2 capture, transport, and storage across the five system designs described in the text involving all 22 facilities in Cluster 4. Capture and transport costs are amortized over 30 years, assuming a weighted average cost of capital of 10%/y. Levelized storage cost is assumed to be $10/tCO$_2$ (as discussed in Section 4.2). Squares and triangles indicate aggregate kilometers of CO_2 pipelines and flue gas ducts, respectively.

calculated from

$$C_K = c_K \times D_{out} \times L \tag{15}$$

where c_K is construction cost per external wall area. See SI.11 for all parameter values for these calculations.

4.3. CO_2 capture, transportation, and storage system designs

Combining the CO_2 capture, transport, and storage models described above, we explore here five different system designs for Cluster 4. For each design, *SimCCS*^{PRO} determines the pipeline and/or flue-gas duct routings that minimize total levelized cost of CO_2 capture, transport, and storage. Fig. 12 shows the optimal layouts for the five systems:

1. Individual capture plants at each industrial facility and no sharing of CO_2 pipeline infrastructure delivering CO_2 to the storage sites.
2. Individual capture plants at each industrial facility; intra-sub-cluster sharing of CO_2 pipelines is allowed (inter-sub-cluster sharing is not).
3. Individual capture plants at each industrial facility; unrestricted sharing of CO_2 pipeline infrastructure is allowed within the entire cluster.
4. One capture plant is shared by all facilities within a sub-cluster[1]; intra-sub-cluster sharing of flue gas ducting is allowed between individual facilities; unrestricted sharing of CO_2 pipeline infrastructure is allowed within the entire cluster.
5. One capture plant is shared by all facilities in the cluster*; unrestricted sharing of flue gas ducting and CO_2 pipeline infrastructure is allowed within the entire cluster.

In each of the five designs the collective Cluster 4 emissions are reduced from 8.1 MtCO$_2$/y to 0.37 MtCO$_2$/y (excluding cogeneration emissions, which are discussed below). Fig. 13 reports average levelized costs per tonne of CO_2 captured and stored (assuming Nth-plant costs for capture plants), along with the aggregate kilometers of installed CO_2 pipelines and flue-gas ducting:

- Capture costs dominate total costs and are identical for scenarios 1–3, since there is no sharing of capture plants in those cases. Note that the cluster-average capture cost here masks the range of capture costs for individual facilities (shown in Fig. 11).
- Increased sharing of CO_2 pipeline infrastructure from scenario 1 to 3 reduces transport costs, with a $^2/_3$ reduction from scenario 1 to 3. The reduction in aggregate CO_2 pipeline kilometers (from >700 to about 200) is even larger as a result of scale economies that come with increasing pipeline capacities.
- When shared capture facilities are introduced (scenarios 4 and 5), scale economy benefits in capture costs are observed: average capture costs fall by 19% for scenario 4 and by 36% for scenario 5. These reductions are partially offset by added costs for transport of flue gases, which in scenarios 4 and 5 are 3 and 5 times the level of CO_2 transport costs, respectively.
- For a given scenario, it is possible to roughly estimate the breakeven distance beyond which the economics of ducting CO_2 from an industrial facility to a shared capture plant would be less economically favorable than building a capture plant at the industrial facility. For example, in scenario 5 the savings in total annualized capture cost (compared with scenario 3) is 318 M$/y. The furthest facility from the shared capture site is 10 km (the radius of the cluster analyzed) and the total length of flue gas ducts is 33 km. The annualized ducting cost is 229 M$/y, or an average of 6.9 M$/km-y. Thus, some 13 additional km of ducting (= 318/6.9), or 46 km in total, could be built before ducting costs would exceed the capture cost savings from using a shared capture facility. Assuming the ratio of cluster radius to total ducting length is constant (10/33 for scenario 5), then the cluster radius (i.e., the maximum distance for ducting CO_2 economically) would be about 14 km (= 46 × 10/33). Interestingly, this agrees with the estimate of Berghout et al. [45] for maximum feasible ducting distance.

Our cluster modelling assumed that facilities are not capturing cogeneration CO_2 emissions. In the case of design 5, for example, cogeneration emissions amount to 2.2 MtCO$_2$/y, resulting in a net reduction in cluster-wide emissions of 5.6 MtCO$_2$/y, equivalent to 69% of the original industrial emissions. We did not do detailed modelling with capture of cogeneration emissions but have estimated that the net

[1] Capture plant location is selected manually to be central to the subcluster/cluster and where there are no obvious geospatial conflicts.

reduction in that case would be about 7.6 $MtCO_2/y$ or about 94% of the original emissions. The estimated storage capacity at the two storage sites is sufficient to accept the additional CO_2, so pipeline routings would likely remain as in Fig. 12, but average levelized costs per tonne of CO_2 capture and stored would likely be lower than in Fig. 13 because of cost-scale economies associated with larger capacity capture plants and pipelines.

Finally, emissions arise from the generation of electricity used in conveying CO_2 to storage sites, conveying flue gases to shared capture plants, and boosting CO_2 pressure at the storage site prior to injection. While the grid is expected to be increasingly decarbonized over time, the estimated carbon-intensity of electricity generated in Louisiana at present (2021 data) is about 464 $kgCO_2/MWh$ [91]. Assuming CO_2 pumps used to maintain adequate pressure in CO_2 pipelines use 8.5 kWh/tCO_2 to overcome each 10 bar of pressure drop [131], that flue gas duct fans use 48 kWh/tCO_2 (for typical flue gas with 10% mol CO_2 concentration), and that pumps at injection-well heads raise the CO_2 to close to the estimated average formation's pore pressure [94] (see SI.3), the net reduction in emissions in the case of scenario 5 (as an example) would be 4.9 $MtCO_2/y$, or 61% when cogeneration emissions are not captured. With capture of cogeneration emissions the net reduction would be about 6.7 $MtCO_2/y$, or 83%. In the longer-term, grid decarbonization and use of a low-carbon cogeneration fuel, e.g., hydrogen [10], could reduce or eliminate the "carbon penalty" associated with CO_2 capture, transport, and storage. Our steam-turbine cogeneration design allows for the possibility of switching to a low-carbon fuel in the future, assuming infrastructure for delivering such fuels becomes available.

5. Conclusion

Integrated CO_2 capture systems were designed and simulated for multiple, discreet input CO_2 concentrations and flow rates representing emission streams at existing industrial facilities. Levelized costs of CO_2 capture (LCC) were then estimated using the modeled equipment capacities and cost factors. Functions to calculate LCC for arbitrary CO_2 concentrations and flow rates were derived from the modelling work and embedded in software ($SimCCS^{PRO}$) for optimizing pipeline capacities and routings for transporting CO_2 from capture plants to underground storage formations while minimizing total capture, transport, and storage (CTS) system costs. CTS modelling was developed for a 20-km diameter region in Southeast Louisiana that is currently home to 22 industrial facilities emitting 8.1 $MTCO_2/y$. Cost impacts of various arrangements for sharing of CTS infrastructure were evaluated, ranging from no sharing to maximum sharing of capture and transport systems.

The case study illustrated the utility for scoping studies of the novel cost functions developed in this study, but the functions have limitations. They assume facilities are "retrofit ready", i.e., that facilities have space available and access to other resources needed to build and operate a capture plant retrofit and that target streams need no pre-processing (e.g., for contaminant removal) prior to entering the CO_2 absorber. They also assume existing facilities (along with the capture plant) would ramp-up overnight and operate with 90% capacity factor for 30 years from the start-up year. Thus, the cost functions are not intended to replace detailed engineering studies that would assess facility age, operating conditions, and other factors, but they do enable initial scoping cost estimates with only limited information about target capture streams, namely estimates of CO_2 concentration and flow rate.

Our Louisiana case study estimated costs to build and operate a CTS system involving 22 existing industrial facilities. Without any sharing of CTS infrastructure, the average system cost was 138 $\$/tCO_2$, which would fall by nearly 17% if the 22 facilities maximized sharing of capture and transport infrastructure. Smaller facilities (emitting <1 $MtCO_2/y$, which account for over ¾ of all industrial facilities in Louisiana today) would see substantially larger percentage cost reductions. Importantly, sharing of CTS infrastructure would also reduce by about 2/3 the aggregate length of pipeline needed, which may facilitate shorter permitting and construction times and thereby more rapid decarbonization.

Our work here suggests a few important directions for future studies. (1) The viability of safely pooling CO_2 target streams from multiple industrial facilities via ducting to centralized capture plants warrants further investigation, considering the significant cost benefits identified in our analysis. (2) Future studies might also evaluate the opportunity to transport lean and rich amine solvents to and from shared CO_2 capture plants [45], as an alternative to transporting flue gas in ducts. Case studies in other regions might also explore other CO_2 transportation options, such as trucking or shipping [132]. (3) The design of operating strategies for centralized capture plants that would enable maintaining overall capture performance during unpredicted shutdowns or periodic maintenance of different facilities would be of interest. (4) Business models that could support infrastructure sharing could also be explored. (5) Given the emissions penalty incurred when conventional energy sources are used to meet the parasitic energy needs, exploring the integration of low-carbon options for meeting these energy needs would be worthwhile. (6) Performance modelling here assumed target streams did not require removal of trace contaminants. Future work could include modelling of pre-treatment steps to improve the model. (7) The analysis here considered essentially "overnight" construction of CTS infrastructure. In reality, construction would happen over time. Understanding optimal phasing of CTS build-outs would be informative for planning and investment decision making. (8) The methodology for analyzing CTS systems developed here could be extended to essentially any geographic region, if prospective sites for CO_2 injection are known. In this context, it is of interest to develop a screening approach to identifying potential injection sites in regions of interest. (9) The analysis in this paper considered CO_2 capture and storage as the only industrial decarbonization strategy for an industrial cluster. In reality, a variety of strategies are available (energy efficiency, electrification, hydrogen combustion, and others), and it is of interest to investigate how optimal CTS system designs and costs would change when considered together with other emissions reduction strategies.

CRediT authorship contribution statement

Tubagus Aryandi Gunawan: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hongxi Luo:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation. **Chris Greig:** Writing – review & editing, Validation, Supervision, Project administration, Investigation, Funding acquisition. **Eric Larson:** Writing – review & editing, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data is attached in the supplementary information.

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Center for Energy and the Environment.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apenergy.2024.122775.

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